

AI and Robotics Mastery Roadmap

Phase 1: Programming & Math Foundation

- Python: Data types, control structures, functions, OOP, modules
- Python Libraries: NumPy, Pandas, Matplotlib, Seaborn
- Math: Linear Algebra (vectors, matrices, eigenvalues, transformations)
- Math: Calculus (derivatives, integrals, optimization)
- Math: Probability & Statistics (distributions, Bayes theorem)

Phase 2: Core Machine Learning

- Supervised Learning: Linear/Logistic Regression, Decision Trees, SVMs, KNN
- Unsupervised Learning: Clustering (K-Means, DBSCAN), Dimensionality Reduction (PCA, t-SNE)
- Model Evaluation: Cross-validation, ROC/AUC, confusion matrix, F1-score
- Feature Engineering, Data Cleaning, Pipelines

Phase 3: Deep Learning

- Neural Networks: Perceptron, MLPs, forward/backpropagation
- Activation Functions, Loss Functions
- CNNs: Image processing, filters, pooling, transfer learning
- RNNs/LSTMs/GRUs: Sequence modeling, time-series, text
- Frameworks: TensorFlow, Keras, PyTorch basics

Phase 4: Generative AI

- Transformers Architecture: Self-Attention, Encoder/Decoder blocks
- LLMs: BERT, GPT, T5, LLaMA
- Fine-tuning and using HuggingFace Transformers
- Prompt Engineering & Applications
- Diffusion Models (Stable Diffusion, DALL-E) for image generation

Phase 5: Natural Language Processing (NLP)

- Tokenization, Embeddings (word2vec, GloVe, BERT embeddings)
- NLP Tasks: Text Classification, Summarization, QA, Translation
- Named Entity Recognition, POS tagging
- Chatbot development using Transformers + APIs

Phase 6: Reinforcement Learning

- Q-Learning, SARSA, Policy Gradients, Deep Q Networks
- OpenAI Gym Environments
- Agent-Environment Interactions, Rewards, Exploration vs Exploitation
- Multi-agent systems, self-learning

Phase 7: Robotics & Autonomous Systems

- Arduino + Python Basics (Sensor I/O, Communication protocols)
- Computer Vision with OpenCV (object detection, edge detection, contours)
- Robot Decision-Making using AI models
- Path Planning (A*, Dijkstra) and SLAM
- ROS (Robot Operating System) Basics
- Designing autonomous bots: obstacle avoidance, vision-based navigation

Phase 8: Deployment & Frontend

- Model Deployment with Flask/FastAPI
- Creating Chatbot UI with HTML/CSS/JS
- Connecting Frontend with Backend using REST APIs
- Streamlit and Gradio for AI Demos
- Dockerizing AI apps, Hosting on Cloud (Render, Vercel, HuggingFace Spaces)
- Version Control with Git & GitHub